ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	2	3	2	3	3	3	3	1	3	1	3	2	3	3	1	4	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	2	2	2	2	4	2	4	4	2	2	4	3	2	2	1	3	2	1	2
Q.	41	42	43	44	45															
A.	4	3	3	2	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	4	1	4	3	4	1	2	1	3	1	1	3	2	1	2	1	1	1	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	4	2	3	1	4	4	3	1	4	2	1	2	3	4	2	1	2	1	4
Q.	86	87	88	89	90															
A.	2	2	4	2	3															

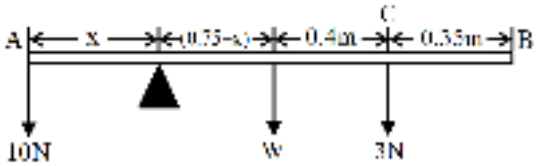
BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	1	3	1	4	2	3	3	1	3	4	3	1	4	4	1	3	4	3	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	4	1	1	1	4	2	2	4	3	1	4	2	4	3	1	4	3	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	2	3	3	2	1	4	3	4	2	1	4	2	1	2	1	2	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	4	1	3	2	4	1	3	4	3	4	4	2	2	4	4	4	3	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	1	2	3	3	2	4	4	2	4										

SOLUTIONS

PHYSICS

1) Let distance of pivot from end A is x



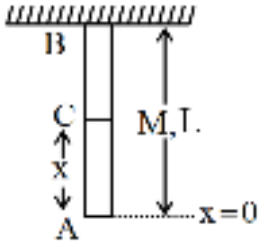
If W = wt. of scale = 0.2 kg × 10 = 2N

From principle of moments : 10x = (0.75 – x) 2 + (1.15 – x)3

10x = 1.50 – 2x + 3.45 – 3x

$\Rightarrow 15x = \frac{4.95}{15} = 0.33m$ = 33 cm

2) Elastic energy density at C

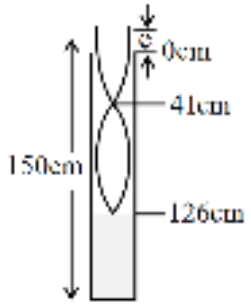


$\Rightarrow u_E = \frac{E}{vol} = \frac{1}{2}\frac{(stress)^2}{Y}$ = constant

$= \frac{1}{2}\frac{\left(\frac{M}{L}\cdot\frac{x}{A}g\right)^2}{Y}$ = constant $\Rightarrow Y \propto x^2$

3) ℓ_1 = 41 cm; ℓ_2 = 126 cm

$\lambda = 2(\ell_2 - \ell_1) = 2(126 - 41) = 1.7$ m



$V = f\lambda = 200 \times 1.7 = 340\ ms^{-1}$

$e = \frac{\ell_2 - 3\ell_1}{2} = \frac{126 - 123}{2} = 1.5cm$

minimum level of liquid at resonance

= 150 – 126 = 24 cm from bottom

4) For half deflection method :

$G = \frac{RS}{R-S} = \frac{(4900)(98)}{4900-98} = 100\Omega$

Full scale current = $I_g = \frac{E}{R+G} = \frac{5}{5000} = 1mA$

Figure of merit (k) = $\frac{I_g}{20} = \frac{1mA}{20} = 50\mu A/division$

5) Real thickness (d_{real}) = $R_3 - R_1$

= 4.115 – 1.215 = 2.900 cm

Apparent thickness (d_{app}) = $R_3 - R_2$

= 4.115 – 2.115 = 2.000 cm

$\mu = \frac{d_{real}}{d_{app}} = \frac{2.900}{2.000} = 1.45$

6) For resistance current remains constant w.r.t. time.

For capacitor, current decays with time for d.c. source

For inductor, it takes some time to establish full current only bulb glows in series with diode for unidirection current.

Both bulb and LED in series glows for unidirection current

7) $U = -\vec{p} \cdot \vec{E}$

• Initial dipole moment : $\vec{p}_i = \left(3\hat{i} + 4\hat{j}\right) \times 10^{-30} C \cdot m$

• Field : $\vec{E} = 4000\,\hat{i}\,N/C$

• Dot product : $\vec{p}_1 \cdot \vec{E} = (3 \times 10^{-30})(4000) = 1.2 \times 10^{-26}$

So, $U_i = -1.2 \times 10^{-26}$ J

• Final dipole moment : $\vec{p}_f = \left(-4\hat{i} + 3\hat{j}\right) \times 10^{-30} C \cdot m$

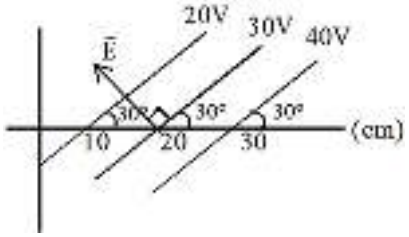
Dot product : $\vec{p}_f \cdot \vec{E} = \left(-4 \times 10^{30}\right)(1000) = -1.6 \times 10^{-26}$

So, $U_f = -(-1.6 \times 10^{-26}) = +1.6 \times 10^{-26}$ J

• Work done by external agent :

$W = U_f - U_i = (1.6 \times 10^{-26}) - (-1.2 \times 10^{-26}) = 2.8 \times 10^{-26}$ J

8)



$E = \frac{\Delta V}{\Delta r} = \frac{(30V - 20V)}{\Delta x \sin 30^\circ}$

$= \frac{10V}{(10cm)\left(\frac{1}{2}\right)} = 200V/m$

Direction :- 90 + 30 = 120

from +x axis.

9) Initially charge on the capacitor

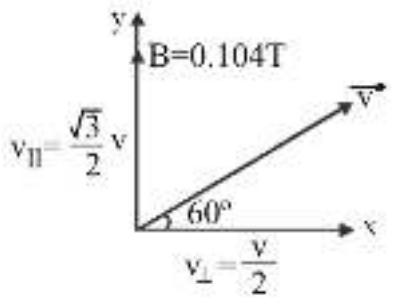
Q = 10×12 = 120 μC

Finally charge on the capacitor
$Q'=(5\times10)\times12 = 600\ \mu C$
So charge supplied by the battery later
$= Q'-Q = 480\ \mu C$

10) $V_a - 6 - 5 + 3 - 6 = V_b$
$V_a - V_b = 14\ V$

11) Resistance of bulb $R = \frac{V^2}{P} \Rightarrow R \propto \frac{1}{P}$
In series combination
$P_{given} = I^2R$
$P_{given} \propto R$
So divided in inverse Ratio of power

12) (i) The field H is dependent of the material core and is
$H = nI = 1000 \times 2.0 = 2 \times 10^3\ Am^{-1}$
(ii) Magnetisation, $M = (B - \mu_0H)/\mu_0$
So, $B = \mu_r\mu_0H$
$= 400 \times 4\pi \times 10^{-7}\ (NA^{-2}) \times 2 \times 10^3\ (Am^{-1})$
$= 1.0\ T$
$I = (B - \mu_0H)/\mu_0$
$= (\mu_r\mu_0H - \mu_0H)/\mu_0$
$= (\mu_r - 1)\ H = 399 \times H = 798 \times 10^3\ Am^{-1}$



13)
• $\vec{v}$ is not parallel to $\vec{B}$
• path of the proton is helical radius
$= \frac{mv_{\perp}}{qB} = 0.1m$, $T = \frac{2\pi m}{qB} = 2\pi \times 10^{-7}$ second.

14) $B = \frac{\mu_0 nI}{2r}$
here $\phi = 2\pi rn$ = constant
$\Rightarrow r = \frac{\ell}{2\pi n} \Rightarrow B = \frac{\pi\mu_0 n^2 I}{\ell}$
$\Rightarrow B \propto n^2$

15) $p_{in} = \frac{880}{88} \times 100$

$= 1000$ W

$I_{input} = \frac{1000}{2200}$

$= 0.4545$ A

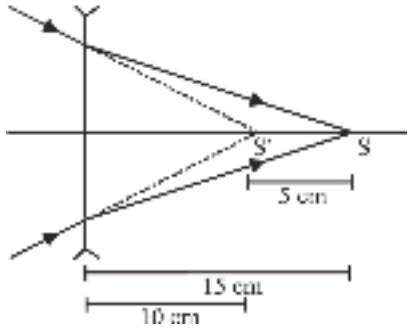
16)

$Q = \frac{W_r L}{R} = \frac{1}{\sqrt{LC}} \cdot \frac{L}{R}$

$= \frac{1}{R}\sqrt{\frac{L}{C}}$

$= \frac{1}{3} \times \sqrt{\frac{1}{9}} = \frac{1}{3 \times 3} = \frac{1}{9}$

17)

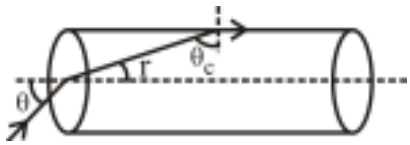


$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

u = 10, V = 15, f = ?

$\frac{1}{15} - \frac{1}{10} = \frac{1}{f} \Rightarrow f = -30$ cm

18)



$\theta_c = \sin^{-1}\frac{1}{2/\sqrt{3}}$

$\theta_c = 60°$

$r = 30°$

$1 \sin\theta = \mu \sin r$

$\sin\theta = \frac{2}{\sqrt{3}} \times \frac{1}{2}$

$\theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

19) Imax = 4 I

$\frac{I_{max}}{4} = I + I + 2\sqrt{II}\cos\Delta\phi$

$I = I + I + 2.I\cos\Delta\phi$

$\Delta\phi = \frac{2\pi}{3}$

$\frac{2\pi}{\lambda}(\Delta x) = \frac{2\lambda}{3}$

$\Delta x = \lambda/3$

$d \sin\theta = \frac{\lambda}{3}$

$\sin\theta = \frac{\lambda}{3d}$

20)

$\phi = \phi_0 + eV_0$

$\frac{12400}{4000} eV = \phi_0 + e(1.4V)$

$3.1\ eV = \phi_0 + 1.4\ eV$

$\phi_0 = 3.1\ eV - 1.4\ eV$

$\phi_0 = 1.7\ eV$

21)

8.25×10^{-11} m

22)

$P = \frac{N\in_1}{t} \Rightarrow 2 \times 10^3$

$= \left(\frac{N}{t}\right) 200 \times 10^6 \times 1.6 \times 10^{-19}$

$\frac{N}{t} = 6.25 \times 10^{13}$

23)

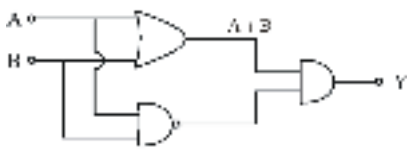
$L = \frac{nh}{2\pi} = \frac{3h}{2\pi}$

then n = 3

$K.E. = \frac{13.6}{n^2} eV = \frac{13.6}{9}$

$= 1.51\ eV$

Final answer : Option (2)



24)

$\overline{A.B}$

$= \overline{A} + \overline{B}$

$= (A+B)\left(\overline{A}+\overline{B}\right)$

$= A\bar{A} + A\bar{B} + B\bar{A} + B\bar{B}$

$= A\bar{B} + B\bar{A}$

= EX–OR

OR

A B $\rightarrow G_1 \rightarrow \overline{A+B}$

A B $\rightarrow G_2 \rightarrow \overline{A.B}$

G_1 G_2 $\rightarrow G_3$ $\overline{A+B} + \overline{A.B}$

$\overline{A}.\overline{B} + \overline{A.B}$

$A\overline{B} + \overline{A}B$

25)

Assertion (A) states that adding an external charge doesn't change the flux through the Gaussian surface but changes the electric field E on the surface. This is correct. Gauss's law states that the flux depends only on the enclosed charge (Q_{in}), which remains unchanged. However, the electric field E at each point on the Gaussian surface is the vector sum of the fields due to all charges (inside and outside), so it will change when an external charge is added.

Reason (R) states that the electric field E at any point on the Gaussian surface is only due to the inside charge. This is incorrect. The electric field at any point in space is the superposition of the electric fields due to all charges present, both inside and outside the Gaussian surface.

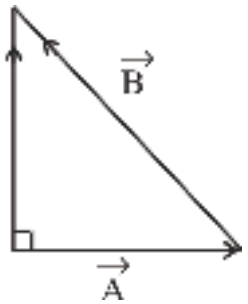
Therefore, Assertion (A) is correct, but Reason (R) is incorrect.

26)

Bird sits on a single wire, its whole body is at same potential. For current flow there should be potential difference.

27) $\alpha = 90°$

$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$



$$\cos\theta = \frac{-A}{B} \quad \text{.....(i)}$$

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

$A^2 = A^2 + B^2 + 2AB\ (-A/B)$

$2A^2 = B^2 \Rightarrow \frac{A}{B} = \frac{1}{\sqrt{2}}$ .....(ii)

from (i) and (ii)

$$\cos\theta = \frac{-1}{\sqrt{2}}$$

$$\theta = \frac{3\pi}{4}$$

28)

$m \propto v^x \rho^y g^z$

$[m] = [LT^{-1}]^x [ML^{-3}]^y [LT^{-2}]^z$

On solving x = 6, y = 1, z = –3

29)

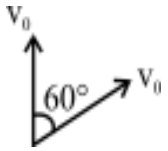
Slope of curve $\frac{d}{dx}v^2 = \frac{12-0}{2-0}$

$= 2v\frac{dv}{dx} = 6$

$v\frac{dv}{dx} = 3$

$a = v\frac{dv}{dx} = 3\ m/s^2$

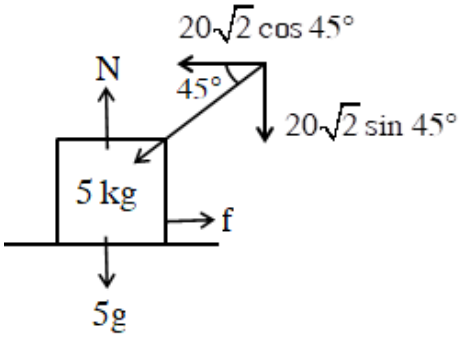
30)



$v_{rel} = v_0$

$d = v_{rel \times t}$

31)



$N = 5g + 20\sqrt{2}\sin 45° = 70$

$f_k = \mu_k N$

$= \frac{1}{2} \times 70 = 35$

$F_{net} = 35 + 20$

$5a = 55$

$a = 11\ m/s^2$

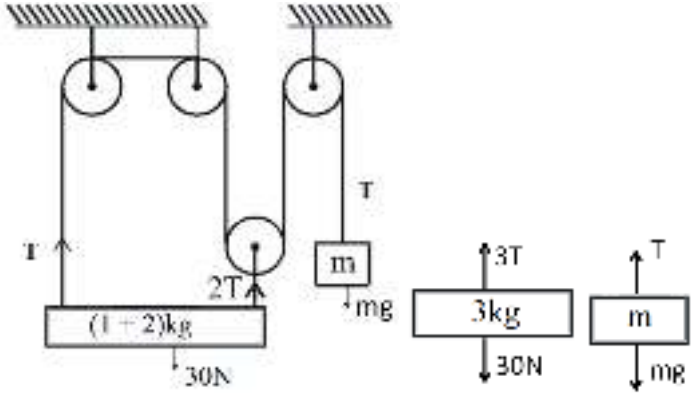
$V = u - at$

$V = 33 - 11 \times 3$

V = 0

32)

$F_{net} = 0$



3T = 30 .... (1)
T = mg .... (2)
Put T in equation ....(1)
3mg = 30
m = 1kg

33) $\frac{P'}{P} = \left(\frac{v'}{v}\right)^3$

Tripling the water flow rate means tripling the speed (v′ = 3v).

$\frac{P'}{P} = \left(\frac{3v}{v}\right)^3 = 27$

P′ = 27 P

34)

$N = mg + \frac{mv^2}{r}$

$r_A < r_B$

$\Rightarrow N_A > N_B$

35)

After collision,
Speed of 1st particle is

$\vec{v}_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)\vec{u}_1 + \vec{0}$

$v_1 = \left(\frac{m - 3m}{4m}\right)v_0 = -\frac{v_0}{2} = \frac{v_0}{2}(ACW)$

Speed of 2nd particle is

$\vec{v}_2 = \frac{2m_1}{m_1 + m_2}\vec{u}_1 + \vec{0}$

$v_2 = \frac{2m}{4m}v_0 = \frac{v_0}{2} = \frac{v_0}{2}\ (CW)$

Time after which next collision takes place is

$$t = \frac{S_{rel}}{V_{rel}} = \frac{2\pi R}{\left(\frac{v_0}{2} + \frac{v_0}{2}\right)} = \frac{2\pi R}{v_0}$$

$$t = \frac{2\pi R}{v_0}$$

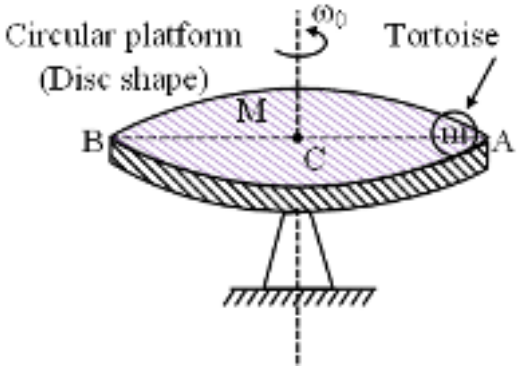
36) $T_1 + T_2 = 2Mg$ .....(1)

torque about 'c'

$$T_1.\frac{L}{2} + Mg.\frac{L}{4} = T_2.\frac{L}{2}$$

$2T_1 + Mg = 2T_2$ .....(2) on solving (1) and (2)

$T_2 = \frac{5Mg}{4}$ and $T_1 = \frac{3Mg}{4}$



37)

If the tortoise moves along A ⊂ B on this rotating circular platform. Then what effect will be there on the angular speed of the platform with time.

This question is based on conservation of angular momentum.

$L = I\omega$ = constant

For the motion of tortoise from point A to C →

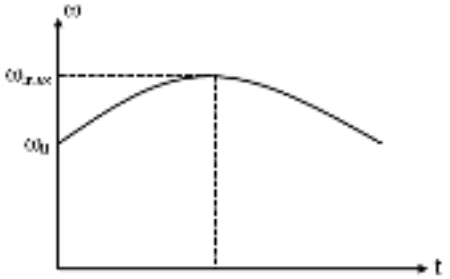
$I_{system} \downarrow \Rightarrow \omega_{system} \uparrow$

For the motion of tortoise from point C to B →

$I_{system} \uparrow \Rightarrow \omega_{system} \downarrow$

So, we can say that angular velocity of the disc first increases, then becomes maximum & then decreases.

So, out of all the "ω vs t" graphs given in the options, the best suited graph will be -



$$\frac{t_1}{t_2} = \frac{\sqrt{H} - \sqrt{\frac{H}{4}}}{\sqrt{\frac{H}{4}} - 0} = 1$$

38)

39)

$W + F_V = Th$

$F_V = Th - W$

but $\sigma = 4\rho$ (density of liquid is 4 times as that of the material)

$\Rightarrow \quad F_V = V(4\rho)g - V\rho g$

or $|F_v| = 3V\rho g = 3W \Rightarrow \frac{F_v}{W} = 3:1$

40) $W = P\Delta V = P(V_s - V_w) = 10^5(1521-1) \times 10^{-6}$ J

$= 152$ J

and $Q = 1 \times 540 \times 4.2$ J $= 2268$ J

$\Delta U = Q - W = 2268 - 152 = 2116$ J ≈ 2.12 kJ

41)

Wein's displacement law $\lambda_m = \frac{b}{T}$

b → Wein's constant

T → absolute temperature

λ_m = wavelength corresponding to maximum spectral intensity

42) Heat gain by Ice = Heat loss by water

$-20^\circ C \rightarrow 0^\circ C \quad 0^\circ C \rightarrow 10^\circ C$

Ice Ice water water

$m_i S_i \Delta Q_i = m_w S_w \Delta Q_w$

$10 \times \frac{1}{2} \times 20 = 10 \times 1 \times 10$

Final condition ⇒ 0°C Ice of 10g

+

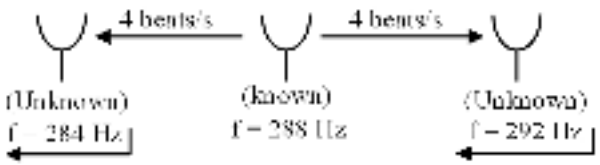
0°C water of 10g

43) Let the mass be displaced by a distance x, from equilibrium.

⇒ Net force on mass M is 2 (2 kx) = 4 kx

$\Rightarrow a = -\frac{4kx}{M} \Rightarrow \omega = \sqrt{\frac{4k}{M}}$

$= \sqrt{\frac{4 \times 100}{25}} = 4$ rad/s



44)

= 283 Hz = 291 Hz

= 282 Hz = 290 Hz

= 281 Hz

$\therefore F_{unknown} = 292$ Hz (initial)

45)

Given wave equations:
$y_1 = 2a \sin(\omega t - kx)$, $y_2 = a \sin(2\omega t - 2kx)$
From the intensity formula:
For y_1:
$I_1 \propto (2a)^2\omega^2 = 4a^2\omega^2$
For y_2 :
$I_2 \propto (a)^2(2\omega)^2 = a^2 \times 4\omega^2 = 4a^2\omega^2$
Since $I_1 = I_2$, Statement-1 is true.
Statement-2:
It states that intensity depends only on amplitude, but we just saw that frequency also affects intensity.
Since $I \propto A^2\omega^2$, not just A^2, Statement-2 is false.

CHEMISTRY

46)

Z = 29 → electronic configuration: [Ar] $3d^{10}$ $4s^1$.
(a) n = 4, ℓ = 0, m = 0 → orbital 4s contains 1 electron → 1.
(b) n = 3, s = +1/2 → n = 3 has $3s^2$, $3p^6$, $3d^{10}$ = 18 electrons; half with s = +1/2 → 9.
(c) Penultimate shell = n = 3 → 18 electrons.
(d) Kernel = all electrons except outermost shell (n = 4) → 29 − 1 = 28.
Final Answer : Option (3)

47)

$Cl_2(g) + 3F_2(g) \rightleftharpoons 2ClF_3(g)$, $\Delta H = -329$ kJ

Removing Cl_2
→ Decreases reactant concentration ⇒ shifts left

A. Increasing temperature
   → Exothermic reaction ($\Delta H < 0$), so adding heat favors reverse direction
B. Adding inert gas at constant volume
   → No change in partial pressures ⇒ no effect on equilibrium
C. Decreasing volume of the container
   → Increases pressure
   → Gas moles: Left = 1 (Cl_2) + 3 (F_2) = 4 mol
   Right = 2 (ClF_3) = 2 mol
   → Equilibrium shifts toward fewer moles, i.e., to the right
   → More ClF_3 will form

48)

$n_{CO} = \frac{14}{28} = 0.5$ mol
Volume at STP = 0.5 × 22.4 = 11.2 L
No. of molecules = $0.5 \times 6.023 \times 10^{23} = 3.011 \times 10^{23}$

49)

$CaCO_3 \rightarrow CaO + CO_2$
100g 22.4 L
22.4 L CO_2 is produced by = 100 g $CaCO_3$
1 L CO_2 is produced by = $\frac{100}{22.4}$
44.8 L CO_2 is produced by = $\frac{100}{22.4} \times 44.8$
= 200 g
$CaCO_3$ is 80% pure
$x \times \frac{80}{100} = 200$
$x = \frac{200 \times 100}{80} = 250\,g$

50) $\overset{+1}{Li}\overset{-1}{H} \rightarrow \overset{0}{Li} + \frac{1}{2}\overset{0}{H_2}$
Intramolecular redox reaction.

51)

$2Al(s) + 3Fe^{+2} \rightarrow 2Al^{+3} + 3Fe(s)$
$E^{o}_{Cell} = -0.44 - (-1.66) = 1.22$ V
From nernst's equation
$E_{cell} = 1.22 - \frac{0.06}{6} \log_{10} \frac{(0.2)^2}{(0.01)^3}$
$= 1.22 - 0.01 \times \log 4 \times 10^4$
$= 1.22 - 0.046$
or $E_{cell} = 1.174$

52)

$t_{1/2} \propto (a)^{1-n}$
if n = 0 then $t_{1/2} \propto a$

53) $\left(\frac{\omega}{E}\right)_{H_2} = \frac{965 \times 10}{96500} = 0.1$
$\left(V_{H_2}\right)$ at NTP = 0.1 × 11.2 L = 1.12 L

54)

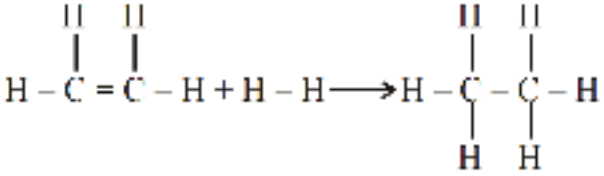
Boiling point elevation ($\Delta T_b = i \times K_b \times m$) depends on the van't Hoff factor (i)
• Urea (non-electrolyte): i = 1
• KCl (electrolyte $K^+ + Cl^-$) i = 2
since $i_{KCl} > i_{urea}$, the ΔT_b for KCl is greater. Thus 0.1 M urea has a lower boiling point. Assertion (A) is correct. The reason (R) correctly defines this colligative property. Reason (R) is correct and explains (A).

55)

For ideal solutions:
$P_A = x_A P_A^o$
$P_A = 32, P_A^o = 40$
So,
$x_A = \frac{32}{40} = 0.8$
Thus,
$x_B = 1 - x_A = 1 - 0.8 = 0.2$

56) $\Delta G = \Delta H - T\Delta S$
For spontaneous ΔG = –ve
$\Delta H - T\Delta S < 0$
$\Delta H < T\Delta S$
$T > \frac{\Delta H}{\Delta S}$ $T > \frac{400}{0.2}$ T > 2000

57)



ΔH reaction = ΣB.E. (Rectant) – ΣB.E. (product)
= [B.E.(4C – H) + B.E. (C = C) + B.E. (C – C)]
– [B.E.6(C – H) + B.E.(C – C)]
= B.E. (C = C) + B.E. (H – H) – 2 B.E.(C – H) – B.E. (C – C)
= 600 + 400 – 2 × 410 – 350 kJ
= – 170 kJ

58)

From Arrhenius equation
$\ln K = \ln A - \frac{E_a}{RT}$
 slope of the curve in between
lnK and 1/T is = $-\frac{E_a}{R}$
$\Rightarrow -\frac{E_a}{R} = -4.8 \times 10^3$

Or $E_a = 4.8 \times 10^3 \times 8.314$
$= 39.9$ KJ mol^{-1}

59)

$[H^+] = \sqrt{K_a \times C}$
given pH = 3
so $[H^+] = 10^{-3}$
$10^{-3} = \sqrt{K_a \times 10^{-1}}$
$\sqrt{K_a} = \frac{10^{-3}}{10^{-1/2}} = 10^{-5/2}$
$K_a = 10^{-5}$
Final answer : 2

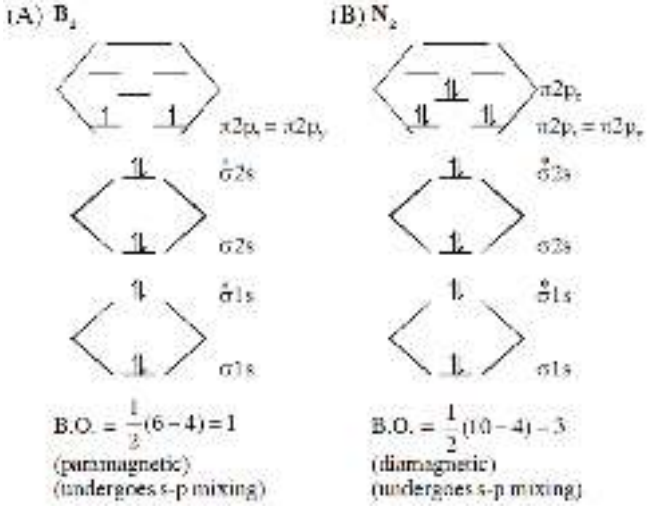
60)

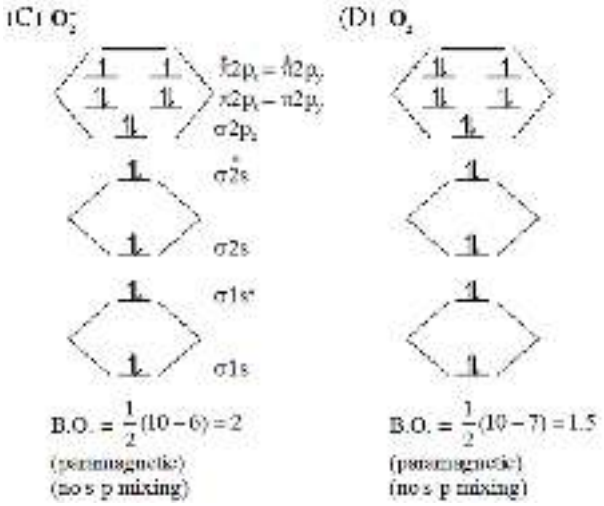
In titration of weak acid and strong base at end point the pH change will be in the range of 7 – 11, therefore suitable indicator is phenolphthalein.

61) NCERT-11th Pg # 85

62) SiO_2, Cl_2O_7, B_2O_3, CO_2, SO_2 are acidic oxide

63) s–p intermixing is operative in molecule having upto $14e^-$.

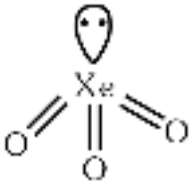




All B_2, N_2, O_2, O_2^- can either gain or loss electron.
 Can undergoes oxidation as well as reduction.

64)

(A) $H_2O > NH_3 > CHCl_3$ – correct



(B) 1 LP
XeF_4 XeO_3 XeF_2 – Incorrect
(2 LP) (1 LP) (3 LP)
(C) O–H > C–H > N–O – Incorrect
According to size of element.
(D) N_2 > O_2 > H_2 – correct
N≡N O=O H–H

65) $XeF_3^+ \rightarrow sp^3d$ (T shape), $XeF_5^\ominus \rightarrow sp^3d^3$
$XeO_2F_2 \rightarrow sp^3d$ (see saw), $XeO_3 \rightarrow sp^3$ (pyramidal)

66) $\text{L.E.} = \frac{\text{Charge}}{\text{Size}}$

67)

$PtCl_4.3NH_3 \Rightarrow [Pt(NH_3)_3Cl_3]Cl$
It is heteroleptic and show fac-mer isomerism

68) Question Explain: Match the Column.
Concept:
Solution: $Cu^+ = sp^3$(C.N = 4)

$C_2O_4^{-2}$ with Co^{+3} acts as SFL $Pt\overline{Cl}_4^{-2}$ = dsp^2

69) (ii) $[Ti(H_2O)_6]^{3+}$ shows voilet colour due to d-d tranistions but when it heated H_2O molecules which are acting as ligands are vapourised. Hence in the absence of no any ligand field no splitting occurs.
(iv) In the presence of strong field ligand more splitting occurs.

70) Chromyl chloride test

71)

Actionids on reaction with boiling water gives oxide and hydride.

72) Fact

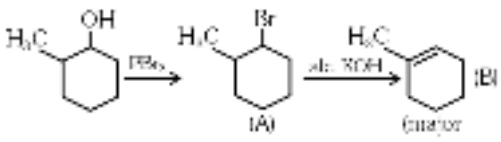
73) **Solution :**
(A) (True) $I_2O_5 + CO \rightarrow I_2 + CO_2$
This reaction is used to **estimate CO** quantitatively.
(B) (True) Chlorine dioxide (ClO_2) is used for **bleaching paper and textiles**.
(C) Stability of oxides
Correct order: I > Cl > Br
(E) (True) Fluorine forms **only HOF** due to **highest electronegativity and absence of d-orbitals.** Other halogens (Cl, Br, I) form multiple oxyacids.
Final Answer : Option (3)

74) NCERT Page No.# 70

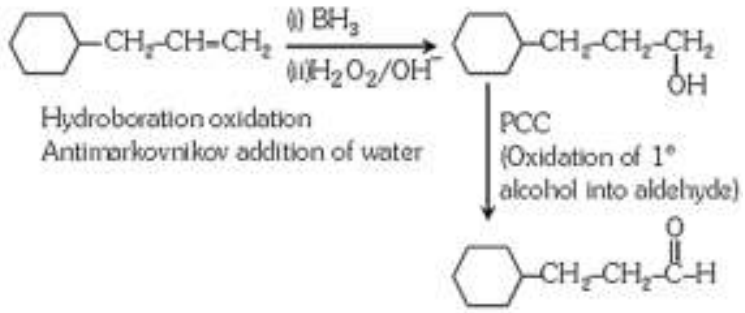
75) NCERT-12 (Old),Chapter-p-Block, Pg. No. 171

76)

due to + M- effect of –C□, C – C□ bond aquires double bond character and does not give phenol.

77) 

78)



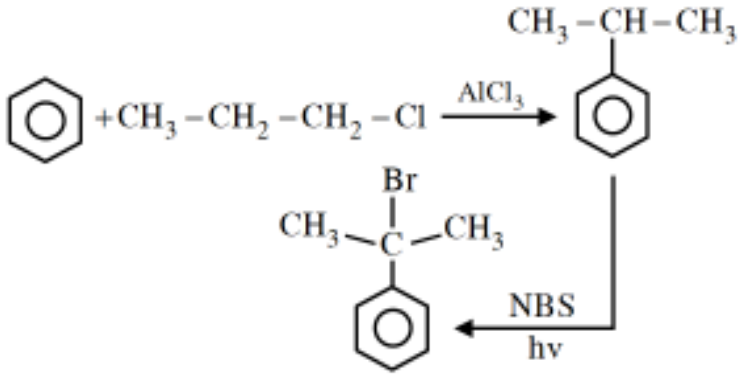
79)

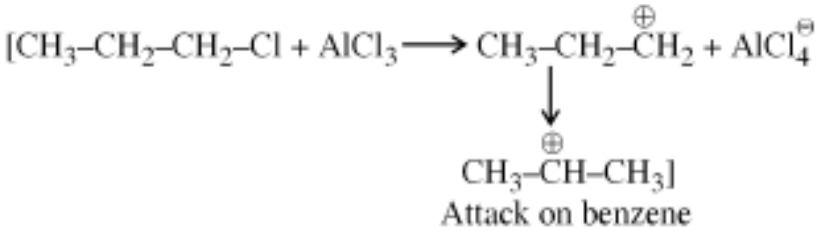
Explaining the Question :- Match the column^{-1} (reactions) with column II (name of reaction).
Concept:- Name reaction
Solution :-

A. fact based.
B. Names of reaction as per the theory.

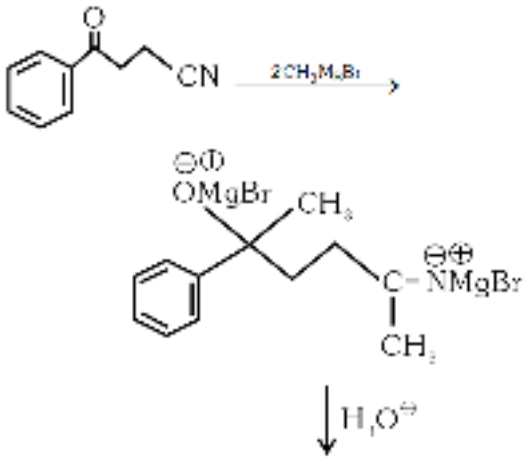
Final Answer :- (3)

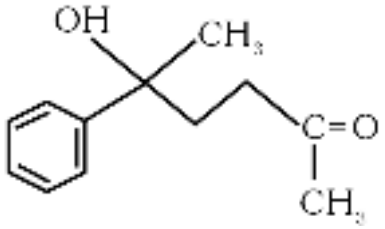
80)



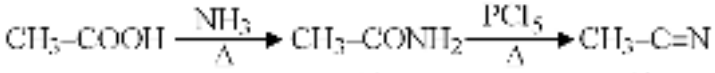


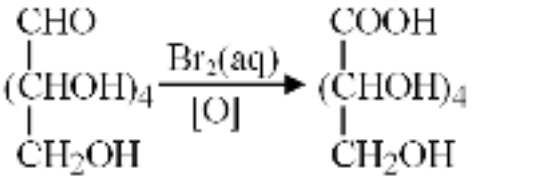
81)





82) If follows Huckel's Rule of aromaticity.

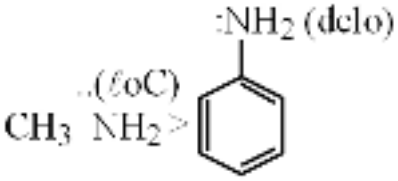
83) 

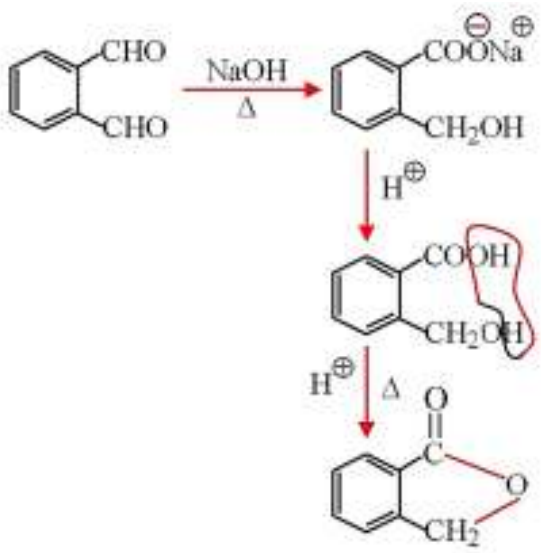
84) 

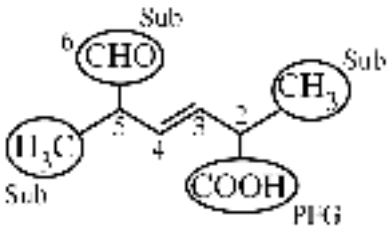
85) Reactivity towards EAR ∝ stability of carbocation

86)

CH_3NH_2 is more basic than aniline



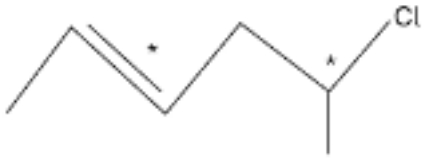
87) 

88) 

2, 5-Dimethyl-6-oxohex-3-enoic acid

89)

Concept:- If molecule is unsymmetrical then
Total no. of stereo isomers = 2^n
Where 'n' is the number of stereockiter (chiral center)
Ans. Exp:-



Stereocenter = 2, Unsymmetrical
Total no. of S.I = $2^n = 2^2$ = (4) isomer
Final Answer: (2)

90)

Under reduced pressure Glycerol is separated from spent lye , because it decomposes before its boiling point.

BIOLOGY

91) NCERT XII, Pg. No. 194

92) NCERT, Pg. # 204, 205

93) NCERT Pg.#173,180,179,182

94)

NCERT-XII, Pg. # 178

95) NCERT Pg.#180,182,184,183

96)

NCERT-XII, Pg. # 183, 185, 187

97)

NCERT Pg. # 40-43

98)

NCERT Pg. # 46

99)

NCERT Pg. # 40

100) NCERT Pg.# 39

101) NCERT_XII_Pg. No. 114

102)

NCERT Pg. # 80

103) **Explaining :** (i) Testes, (ii) Adrenal gland, (iii) Fat bodies, (iv) Cloaca gland.
Concept :
Final Answer : (1)

104)

NCERT XI Pg.# 113

105)

NCERT Pg. # 105

106)

NCERT Pg. # 197

107) NCERT, Pg. # 202

108)

NCERT Pg. # 194

109)

NCERT Pg. # 200

110) NCERT Pg. # 190

111) NCERT Pg. # 189

112) NCERT Pg.# 208, 209

113) NCERT Pg.#211

114) NCERT Pg. 236

115)

NCERT Pg. # 232

116)

NCERT Pg. # 218

117)

NCERT Pg. # 218

118)

NCERT Pg. # 227

119)

NCERT Pg. # 245

120)

NCERT Pg. # 248

121) NCERT, Pg. # 239-250

122) NCERT XI Pg.#245

123) NCERT, Pg # 28

124) NCERT Pg. # 33

125) NCERT(XII) Page#48/54(H) Para:3.5

126) NCERT, Pg. # 27

127) NCERT P.g No. 38

128) NCERT, Pg. # 46

129) NCERT, Pg # 147(E), 158(H)

130) NCERT XII, Pg. # 137, 138

131)

NCERT Pg. # 143

132)

NCERT Pg. # 142

133) NCERT XII Bio P-111

134) NCERT XII Pg. No. # 124, 125

135) NCERT Pg. # 115

136)

Allen module.

137) NCERT, Pg. # 3, 5

138) Its symmetry is icosahedral

139) Three kingdom

140)

Correct Answer (4)

141) NCERT XI, Pg # 27

142) **Explain Question :** The question asks you to match terms related to reproduction in plants, specifically algae, from Column-I with their corresponding definitions or descriptions in Column-II.
Concept : This question is based on asexual and sexual reproduction in lower plants, particularly algae
Explanation :
a) Zoospores
are flagellated (motile) asexual spores. So, (a) matches with (ii).
b) Isogamous
refers to the condition where gametes can be flagellated and are similar in size. So, (b) matches with (iii).
(c) Anisogamous
refers to the fusion of two gametes dissimilar in size. So, (c) matches with (iv).
(d) Oogamous
refers to the fusion between one large non-motile female gamete and a smaller motile male gamete. So, (d) matches with (i).
Final answer : 1

143) NCERT-XI Pg # 67

144) Correct option (2)

145) **Explain Question :**
The image presents a matching question Involving three columns; Column-I (Type), Column-II (Structure), and Column-III (Function), all related to plant tissues. The takk in to correctly natch the items from each column.

Concept :
This question is based on simple permanent tissue.

Explanation :

A. (Parenchyma) matches with a (Structure) and III (Function: Photosynthesis, storage, secretion).
B. (Collenchyma) matches with b (Structure) and i (Function: Mechanical strength and swaying).
C. (Sclerenchyma) matches with c (Structure) and ii (Function: Lignified cell provides fiber).

Final answer : 1

146) Correct Answer (2)

147) NCERT-XI, Pg. # 123, 124, 125

148) NCERT Page No. 138, 139

149)

NCERT Pg. # 121

150)

NCERT XI Pg. No. # 122, 123, 124

151) **Correct Answer: Option 3 – D, B, A, E, C**
Explanation:
Let's break down the order of events during meiosis:
1. D. Chromosome condensation initiates:

A. Chromosomes begin to condense in preparation for meiosis, starting in prophase I.

2. B. Bivalent formation:

A. During prophase I, homologous chromosomes pair up to form bivalents (also called tetrads).

3. A. Crossing over:

A. Crossing over occurs between homologous chromosomes in the bivalent during prophase I (specifically in pachytene), exchanging genetic material.

4. E. Chromosome condensation completes:

A. By the end of prophase I, the chromosomes have fully condensed and become more visible.

5. C. Centromere division:

A. The centromere divides during anaphase II, not earlier in meiosis.

Final Order:

A. D→B→A→E→C

Final Answer:
Option 3-D, B, A, E, C

152) NCERT-XI Pg # 117

153) NCERT (XI) Pg. # 108, table-9.3

154) NCERT Pg # 143, 145, 149

155)

Statement I is correct: In alcohol fermentation, pyruvic acid is converted to ethanol and CO_2 by pyruvate decarboxylase and alcohol dehydrogenase.

Statement II is incorrect: In lactic acid fermentation, lactate dehydrogenase reduces pyruvic acid to lactic acid, but no CO_2 is produced.

156) NCERT-XI Pg # 160

157)

NCERT XI, Pg # 174-175

158)

NCERT XI, Pg # 176

159) NCERT XII, Pg # 10

160) NCERT XII, Pg # 7

161) **Question Explanation :**
The question asks to identify the incorrect statement among the four options provided regarding the structure and development of pollen grains in flowering plants.

Concept:
This question is based on Structure Of Pollen Grain.

Explanation: Generative cell of pollen grain divides meiotically to give rise to the two male gametes. **- Incorrect**
The generative cell divides mitotically, not meiotically, to form two male gametes.

Final answer: Generative cell of pollen grain divides meiotically to give rise to the two male gametes.

162) NCERT XII, Page # 213, 214 (H)

163)

NCERT XII, Page No. # 232

164) NCERT, Pg. # 197

165)

NCERT PG. # 246

166)

NCERT-XII, Pg.# 210

167)

Explain Question: What is Humus?

Concept: Functions of Ecosystem (Decomposition).

Solution:

A dark coloured amorphous substance which is highly resistant to microbial activity is called Humus.

Final answer: 4. Humus.

168) NCERT XII, Pg. # 260, Para-15.1.1, Fig-15.1

169) NCERT (XII) Pg. # 266, para-15.2.2

170)

Birds

171)

Phenylketonuria is autosome linked character.

172)

NCERT-XII Pg. # 88

173)

NCERT-XII Pg. # 67

174)

Question Explanation : match each genetic term in List I with the correct definition/example in List II

Concept : Mendelian Genetics

Solution : A. Two or more alternative forms of a gene → Allele (III)

Matches definition of allele.

B. Cross of F_1 progeny with homozygous recessive parent → Test cross (IV)

Test cross = F_1 × recessive parent.

C. Cross of F_1 progeny with any of the parents → Back cross (I)

Back cross = $F_1 \times (P_1$ or $P_2)$.

D. AB blood group → Co-dominance (II)

Final answer : Option (3) → A–III, B–IV, C–I, D–II

175) Correct Answer: (3)

176) Correct Answer: (2)

177)

Explain Question : The question is asking: To identify which molecular process (like replication, transcription, translation, etc.) occurs at each step in the given sequence — RNA → DNA → DNA → mRNA → Polypeptide.

Concept : This question is based on Central Dogma.

Solution :
The sequence is:
RNA → DNA → DNA → mRNA → Polypeptide

RNA → DNA = Reverse transcription
DNA → DNA = Replication
DNA → mRNA = Transcription
mRNA → Polypeptide = Translation

Final Answer :
A-Reverse transcription, B-Replication,
C-Transcription, D-Translation

178)

NCERT XIIth Pg # 99, 107, 109, 111

179)

NCERT (XII) Pg # 104

180) NCERT Pg # 99